Weekly Report – Topic 11: Expert Labels

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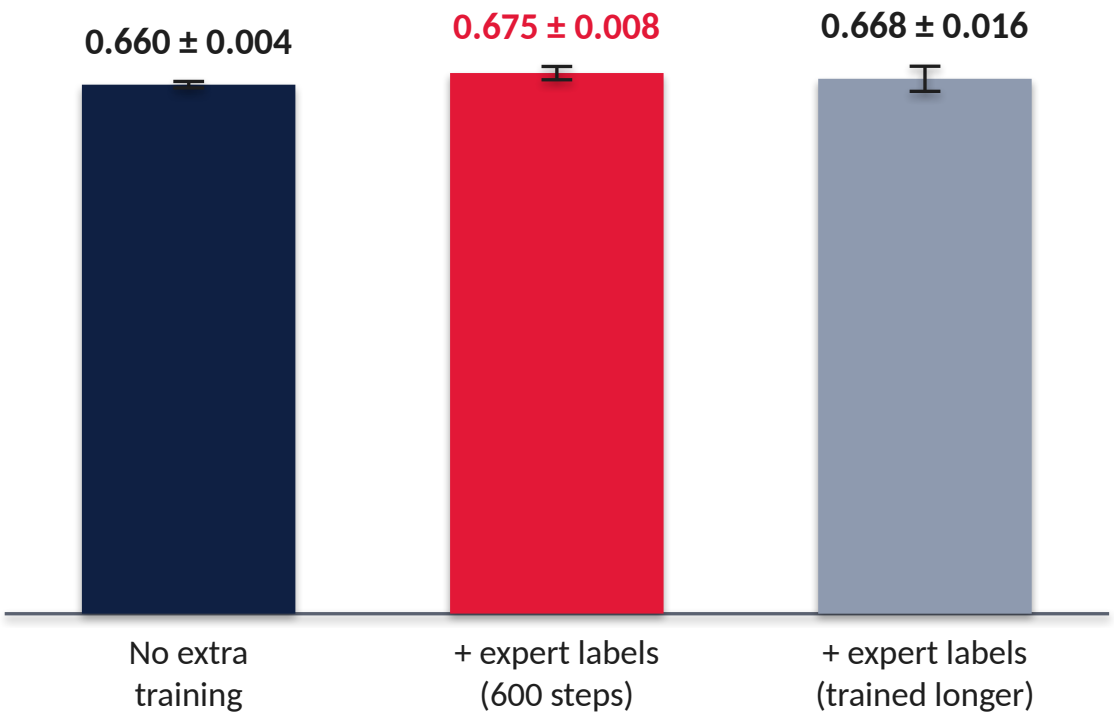
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➢Topic 11: expert labels on our hybrid

What we did

- **The problem:** our hybrid learned from AOFlagger's labels, which agree with the expert only at F1 0.570.
- **Expert labels exist for only 109 images** — the test images.
- **So we split the 109 into 5 parts.** Each part is tested by a model that learned from the other parts' expert labels (68–69 images; 19 more kept for checking).
- **Same hybrid, no changes:** only a short extra training (600 steps).



+0.014 F1 (higher for all 3 seeds, p = 0.03): 0.660 → 0.675 on all 109 images. Level with RFDL's best result with expert labels (0.675), not above it.

max F1, mean ± spread of 3 runs. Results with expert labels must be compared with other expert-label results (RFDL: 0.669 / 0.675), not with the 0.660 table.

➢Which patches should the expert label?

Patches labelled (600 steps)	5%	10%	20%
Where the model is most uncertain	0.6527	0.6551	0.6604
Where the model sees most RFI	0.6572	0.6514	0.6630
At random	0.6667	0.6664	0.6685
Whole images at random	0.6654	0.6568	0.6629

max F1 on all 109 images, mean of 3 runs. No extra training: 0.660 · all expert labels: 0.675.
Share of the label pool: 5% ≈ 3 images, 20% ≈ 14 images.

The "smart" choices lose

Uncertain and most-RFI patches did worse than random ones in **17 of 18** tests.

Why: they are full of RFI — 3.2–7.4% of their pixels, against 0.76% overall. The model learns "RFI is common" and flags too much.

Train longer?

No: worse in **12 of 13** settings. 19 checking images reward lucky checkpoints.

Rule of thumb: train about 600 steps; if the expert's time is short, label random patches.

➢To Do Next

- **Dice-off run (3 seeds, ~2 h):** is the Dice loss what puts our hybrid ahead of tf_unet?
- **A possible fix for patch choice:** keep AOFlagger's labels outside the chosen patches, so the model still sees normal background (not tested yet).
- **Report:** new section 5.6 with these results.